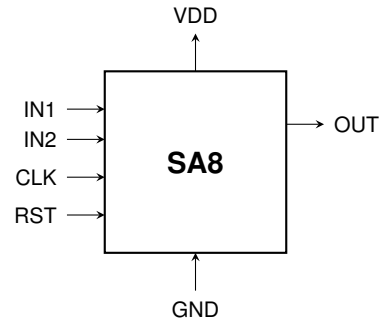


SA8

SISO 8 BIT ADDER

Design Overview

This device contains a full-custom, transistor-level 8-bit Serial Adder with a single serial output and a asynchronous reset. Two 8-bit operands are loaded serially into two 8-bit Input SISO registers and processed through a 1-bit Full Adder over 17 clock cycles, producing a 9-bit result (8 sum bits + carry-out). The combinational datapath consists of a 1-bit Full Adder and a carry D Flip-Flop.



Pin Configuration

Pin Name	I/O Type	Description
VDD	Power	Core supply voltage (1.8V nominal)
GND	Power	Circuit ground reference (0.0V)
IN1	Input	Serial data input 1 (LSB-first)
IN2	Input	Serial data input 2 (LSB-first)
CLK	Input	System Clock (positive edge-triggered)
RST	Input	Asynchronous Reset (Active-Low)
OUT	Output	9-bit serial result output

Operating Mode Table

RST	CLK	OUT	Operating Mode
L	X	–	Hold/Inactive
H (cycles 1–8)	↑	–	Input Loading
H (cycles 8–16)	↑	–	Addition / SUM Generation
H (cycles 17–25)	↑	Serial 9-bit(upto) out	Output Shifting

H = HIGH Logic Level · L = LOW Logic Level · ↑ = Positive-edge transition

Key Specifications Box

Parameter	Value
Technology	180nm
Total Clock Cycles	25
Output Width	9-bit (8 SUM + COUT)
Max Frequency (fmax)	3.85 GHz
Average Power (Random)	78 μ W
Total Energy per Addition	24.375 pJ
Layout Area	200 × 47.5 μ m (9,500 μ m ²)